INCH-POUND

MIL-R-39016/34E
22 June 1992
SUPERSEDING
MIL-R-39016/34D
20 July 1988

MILITARY SPECIFICATION SHEET

(E) RELAYS, ELECTROMAGNETIC, DPDT, LOW LEVEL TO 1 0 AMPERE, (0.100-INCH TERMINAL SPACING)

(E) Inactive for new design after 22 June 1992

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016

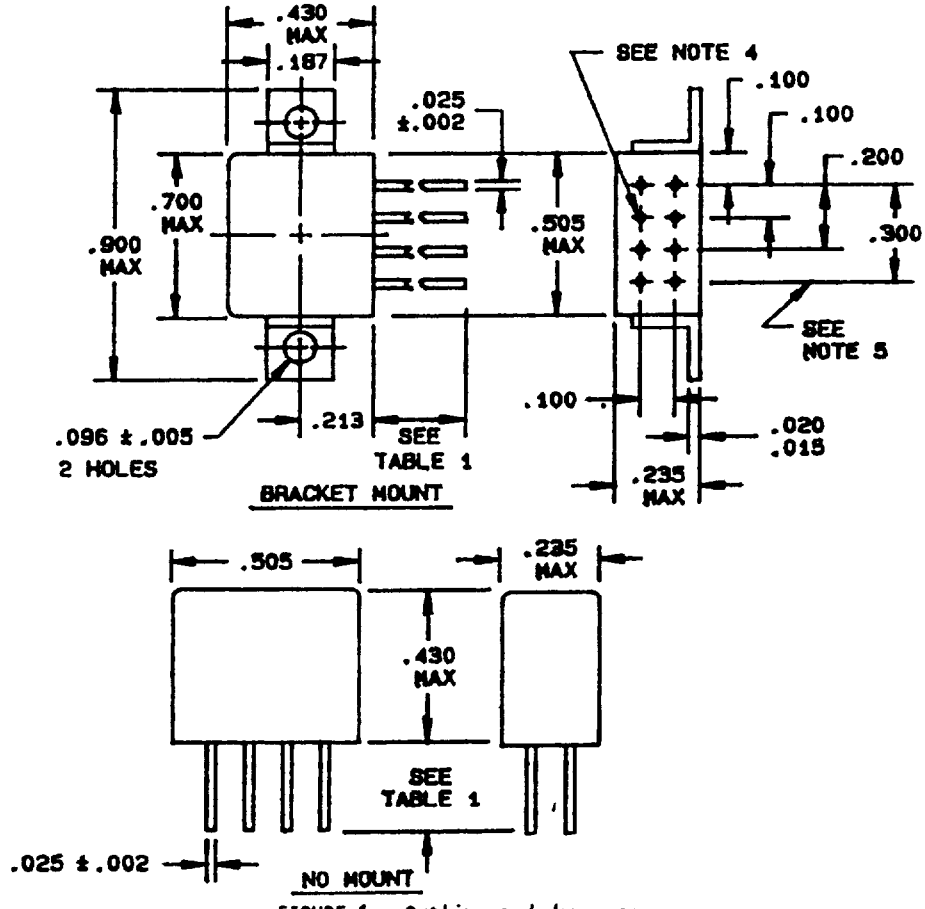


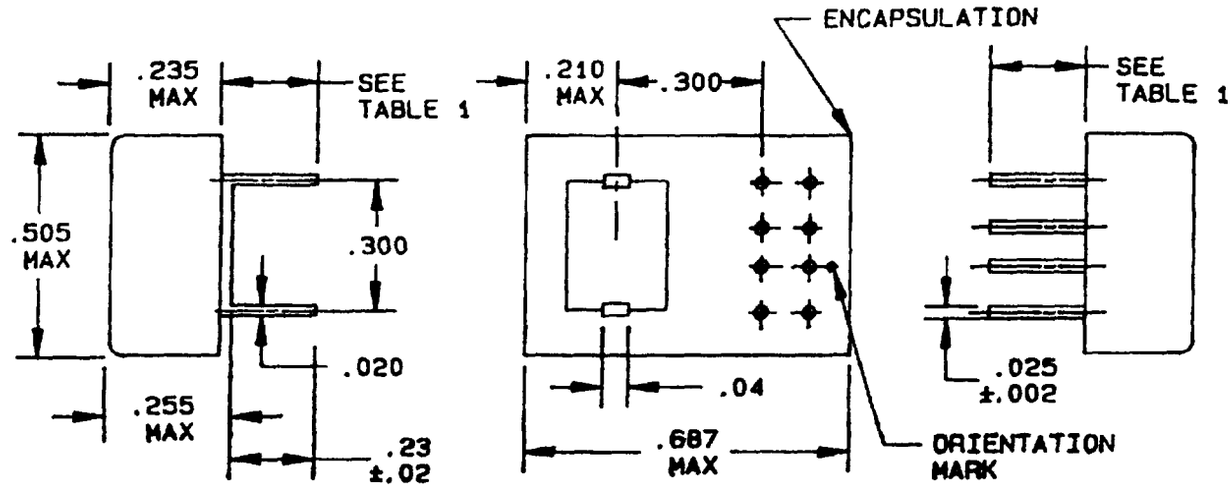
FIGURE 1 Outline and dimensions.

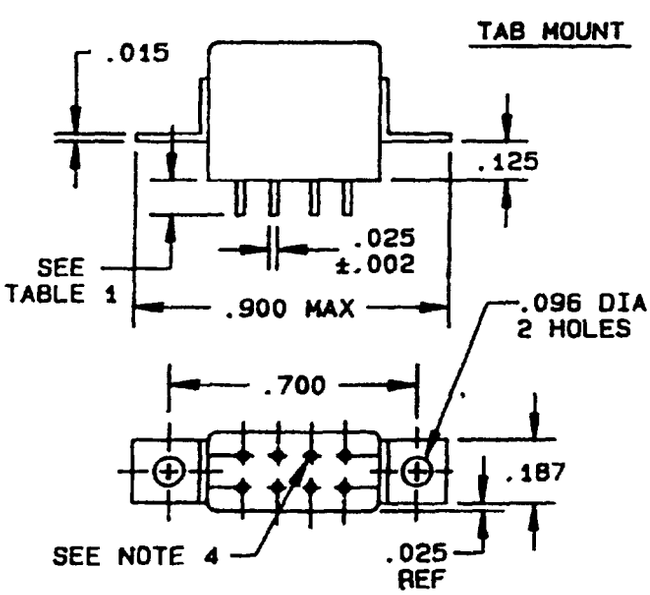
(E) denotes changes

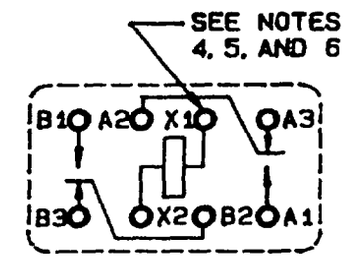
AMSC N/A FSC 5945

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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Inches	mm	Inches	mm
.002	0.05	.213	5.31
.003	0.13	.23	5.8
.015	0.38	.235	5.97
.020	0.51	.25	6.35
.025	0.64	.255	6.48
.04	1.0	.300	7.62
.096	2 44	.430	10 92
.100	2.54	.505	12.83
.187	4 75	.687	17.45
.200	5.08	.700	17.78
.210	5.33	.900	22.86
		1.470	37.34

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm)
4. The X1 terminal shall be identified with a contrasting bead
5. Terminal locations shown are applicable to all type mounts.
6. Terminal numbers shown on circuit diagram are for reference only Numbers do not appear on relay.

FIGURE 1 Outline and dimensions - Continued.

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(E) TABLE I. Dash numbers and characteristics.

Dash numbers				Mount	Coil voltage 4/ (V dc)		At +25°C				Over temperature range		
Wire lead 1 50 ± 020	Wire lead (solder pin-SP) .187 ±.020	Wire lead (solder pin-SP) .25 ±.02	Pin (plug in) .187 ±.020 2/ 3/		Rated	Max	Coil resistance Ohms ±10%	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified drop-out value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)
-001				Bracket	6.0	8.0	90	3.0	1.5	0.3	4.0	2 0	0 2
	-002			No mount	6.0	8.0	90	3.0	1.5	0.3	4.0	2 0	0.2
		-003		Tab mount	6.0	8.0	90	3.0	1.5	0.3	4.0	2 0	0 2
-004				Bracket	12	16	310	6.0	3.0	0.6	8 0	4 0	0.4
	-005			No mount	12	16	310	6.0	3.0	0.6	8 0	4 0	0.4
		-006		Tab mount	12	16	310	6.0	3.0	0.6	8 0	4 0	0.4
-007				Bracket	26.5	32	1500	13 5	10	1.5	18	14	1 0
	-008			No mount	26.5	32	1500	13.5	10	1.5	18	14	1.0
		-009		Tab mount	26.5	32	1500	13 5	10	1 5	18	14	1 0
			-010	No mount	6.0	8 0	90	3.0	1.5	0.3	4.0	2.0	0.2

See footnotes at end of table.

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(E) TABLE I. Dash numbers and characteristics - Continued.

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			-011	No mount	12	16	310	6.0	3.0	0 6	8 0	4 0	0.4
			-012	No mount	26.5	32	1500	13 5	10	1 5	18	14	1 0
			-013	Flange mount	6.0	8.0	90	3 0	1.5	0.3	4 0	2.0	0.2
			-014	Flange mount	12	16	310	6 0	3.0	0 6	8 0	4.0	0.4
			-015	Flange mount	26.5	32	1500	13 5	10	1.5	18	14	1.0

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc or peak ac open circuit are not recommended for subsequent use in low level applications.
2/ Pins (plug-in) are gold plated leads.
3/ Socket plug-in relays are not to be used in Air Force airborne applications.
4/ CAUTION: The use of any coil voltage less than rated coil voltage will compromise the operation of the relay

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REQUIREMENTS

CONTACT DATA:

Load ratings

High level (relay case grounded):

Resistive: 1.0 ampere maximum at 28 V dc maximum

Inductive: 0.2 ampere maximum at 28 V dc maximum with 200 millihenries minimum

Lamp: 0.100 ampere at 28 V dc.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance and voltage drop: 1/

Initial: 0.050 ohm maximum.

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.100 ohm maximum.

Low level.

During life. 33 ohms maximum

After life: 0.150 ohm maximum

Intermediate current:

During intermediate current: 1 ohm maximum.

After intermediate current: 0.100 ohm maximum.

(E) Contact bounce: 2.0 milliseconds (ms) maximum.

(E) Contact stabilization time: 2.5 milliseconds (ms) maximum

Overload (high level only): Two times rated current

COIL DATA: See table I.

Operate time: 4 ms maximum over temperature range with rated coil voltage

Release time 4 ms maximum over temperature range from rated coil voltage

ELETRICAL DATA:

Insulation resistance. 10,000 megohms minimum, except the resistance between coil and case at high temperature shall be 1,000 megohms, or greater. MIL-STD-202, method 302, test condition A

1/ For tab mount only, add 0.025 ohm to the contact resistance values specified

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Dielectric withstanding voltage

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure, and between all contacts in the energized and deenergized positions	500	
Between case, frame, or enclosure and coils	500	300
Between all contacts and coils	500	All terminals to case
Between open contacts in the energized and deenergized positions	350	
Between contact poles	500	

ENVIRONMENTAL DATA

Temperature range -65°C to +125°C.

Vibration (sinusoidal): MIL-STD-202, method 204, test condition D (20 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts (applicable to qualification and group C testing only).

Shock (specified pulse): MIL-STD-202, method 213, test condition C (100 g's) Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts

Magnetic interference. Applicable

Resistance to soldering heat. Applicable

Acceleration Applicable

PHYSICAL:

Terminals: See figure 1

Terminal strength: 1 ± 1 pound pull.

Solderability: Applicable to all terminations except pin, plug-in

Terminal twist test. Applicable to wire leads.

Terminal, socket plug-in: Plug-in terminals shall be gold plated in accordance with MIL-G-45204, type II, class 1 with a nickel underplating that shall be in accordance with QQ-N-290 and 50 to 150 microinches thick.

Dimensions and configuration. See figure 1.

Weight: 0.15 ounce maximum.

Identification marking (full): Applicable, marking surface for date code is optional

(E) LIFE TEST REQUIREMENTS:

Intermediate current. 50,000 cycles

Electrical 100,000 cycles.

Mechanical 1,000,000 cycles

(E) Part or Identifying Number (PIN): M39016/34-(dash number from table I)

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QUALIFICATION INSPECTION

Qualification inspection and sample size: See table II

(E) TABLE II Qualification inspection and sample size

Single submission	Group submission	
50 units plus 1 open unit One failure allowed Qualification inspection as applicable.	M39016/34-007	50 units plus 1 open unit One failure allowed. Qualification inspection as applicable.
	M39016/34-006	2 units, qualification, group II, also shock, vibration, acceleration, terminal strength, seal and resistance to soldering heat.
	M39016/34-013	2 units qualification, inspection group II

SUPERSESSION DATA: See table III

(E) TABLE III. Supersession data.

M39016/34-	M5757/30-
001	025, 028, 031, 034
002	033, 036
004	037, 040, 043, 046
005	045, 048
010	027, 030
011	039, 042
	M5757/19-
007	013, 016, 019, 022
008	021, 024
012	015, 018

(E) QUALITY ASSURANCE PROVISIONS:

Group B not required.
Groups A and C required. Group C testing shall be performed on normal production relays. Lots shall not be generated solely for group C testing, unless requested by the qualifying activity.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

(E) Review activities:
Army - AT, ME, MI
Navy - AS, OS, SH
Air Force - 99
DLA - ES

User activities:
Army - AR, AV

Preparing activity
Navy - EC

Agent:
DLA - ES

(Project 5945-0861)

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